Apache Flink Kickstarter II **[UNDER CONSTRUCTION]**

Apache Flink Kickstarter II is the 2026 evolution of my original Kickstarter project — rebuilt to showcase the cutting edge of **Apache Flink 2.1.x**.

Designed as a hands-on, production-minded accelerator, it brings Flink to life *locally* on **Confluent Platform + Minikube**, while drawing direct comparisons to **Confluent Cloud for Apache Flink** — so you can clearly see what's possible across environments.

Every **example** is delivered end-to-end — from schema design to fully operational streaming pipelines — with implementations in **both Java and Python (when possible)** where it matters, bridging real-world developer workflows with modern streaming architecture.

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1.0 Prerequisites

1.1 Confluent Platform Local Setup

A Makefile-driven quickstart that deploys a full local streaming stack on Minikube:

- **Confluent Platform** (KRaft mode) via Confluent for Kubernetes (CFK)
- **Apache Flink 2.1.1** via the Confluent Flink Kubernetes Operator 1.130
- **Confluent Manager for Apache Flink (CMF) 2.1** for Flink environment management

- **Kafka UI** (Provectus) for cluster inspection

1.1.1 Required Tools

macOS with Homebrew. To install all required tools in one step:

```
make install-prereqs
```

This installs `kubernetes-cli`, `minikube`, `helm`, `gettext`, and `gradle` via Homebrew. Once complete, **launch Docker Desktop** before proceeding.

To verify all tools are present without installing:

```
make check-prereqs
```

> Required: `docker`, `kubectl`, `minikube`, `helm`, `envsubst`, and `gradle`.

1.1.2 Resource Requirements

Minikube is configured with the following defaults, which are required to run the full stack:

Resource	Default
CPUs	6
Memory	20 GB
Disk	50 GB

Override any of these at the command line:

```
make cp-up MINIKUBE_CPUS=8 MINIKUBE_MEM=24576
```

1.1.3 `Makefile` Architecture

```
graph TD
    %% ── Composite entry points
────────────────────────────────────────────
    CP_UP(["`**make cp-up**`"])
    FLINK_UP(["`**make flink-up**`"])
    CP_DOWN(["`**make cp-down**`"])
    FLINK_DOWN(["`**make flink-down**`"])
    TEARDOWN(["`**make confluent-teardown**`"])
```

```
    %% ── Phase 1: Prerequisites ──────────────────────────────────────────
    subgraph P1["Phase 1 — Prerequisites"]
        CHECK_PRE["check-prereqs\ndocker · kubectl · minikube · helm"]
    end

    %% ── Phase 2: Minikube ───────────────────────────────────────────────
    subgraph P2["Phase 2 — Minikube"]
        MK_START["minikube-start\ncpus=6 · mem=20GB · disk=50GB"]
        MK_STOP["minikube-stop"]
    end

    %% ── Phase 3: Confluent Operator ─────────────────────────────────────
    subgraph P3["Phase 3 — Confluent Operator"]
        NS["namespace\nkubectl create namespace confluent"]
        OP_INSTALL["operator-install\nhelm: confluent-for-kubernetes"]
        OP_UNINSTALL["operator-uninstall"]
    end

    %% ── Phase 4: Confluent Platform ─────────────────────────────────────
    subgraph P4["Phase 4 — Confluent Platform"]
        CP_DEPLOY["cp-deploy\nKafka KRaft · SR · Connect\nksqlDB · REST
Proxy · C3"]
        CP_DELETE["cp-delete"]
    end

    %% ── Phase 5: Control Center ─────────────────────────────────────────
    subgraph P5["Phase 5 — Control Center"]
        C3["c3-open\nlocalhost:9021"]
    end

    %% ── Phase 6: Apache Flink ───────────────────────────────────────────
    subgraph P6["Phase 6 — Apache Flink"]
        CERT["flink-cert-manager\ncert-manager v1.18.2"]
        FL_OP["flink-operator-install\nhelm: confluentinc/flink-
kubernetes-operator 1.130.0"]
        FL_RBAC["flink-rbac\nsupplemental Role + RoleBinding\n(services
access for flink SA)"]
        FL_DEPLOY["flink-deploy\nenvsubst → FlinkDeployment CR\ncp-
flink:2.1.1-cp1"]
        FL_UI["flink-ui\nlocalhost:8081"]
        FL_DELETE["flink-delete"]
        FL_OP_UN["flink-operator-uninstall"]
        CERT_UN["cert-manager-uninstall"]
    end

    %% ── Phase 7: CMF ────────────────────────────────────────────────────
    subgraph P7["Phase 7 — Confluent Manager for Apache Flink (CMF)"]
```

```
        CMF_INSTALL["cmf-install\nhelm: confluent-manager-for-apache-flink
2.1.0"]
        CMF_ENV["cmf-env-create\nPOST /cmf/api/v1/environments"]
        CMF_OPEN["cmf-open\nlocalhost:8080/cmf/api/v1/environments"]
        CMF_PROXY["cmf-proxy-inject\nsocat sidecar → C3 Flink tab"]
        CMF_UN["cmf-uninstall"]
    end

    %% ── Phase 8: Kafka UI
──────────────────────────────────────────────
    subgraph P8["Phase 8 — Kafka UI"]
        KUI_INSTALL["kafka-ui-install\nhelm: provectus/kafka-
ui\nbootstrap: kafka:9071"]
        KUI_OPEN["kafka-ui-open\nlocalhost:8080"]
        KUI_UN["kafka-ui-uninstall"]
    end

    %% ── Phase 9: Build & Deploy PTF UDF
──────────────────────────────────
    subgraph P9["Phase 9 — Build & Deploy PTF UDF"]
        BUILD_PTF["build-ptf-udf\n./gradlew clean shadowJar"]
        DEPLOY_CP_PTF["deploy-cp-ptf-udf\ncopy JAR → Flink SQL Client"]
        TEARDOWN_CP_PTF["teardown-cp-ptf-udf"]
        DEPLOY_CC_PTF["deploy-cc-ptf-udf\nJAR → Confluent Cloud"]
        TEARDOWN_CC_PTF["teardown-cc-ptf-udf"]
    end

    %% ── Manifests / Templates
──────────────────────────────────────────
    subgraph FS["k8s/base/"]
        MANIFEST[("flink-basic-deployment.yaml\nFLINK_IMAGE ·
FLINK_VERSION")]
        RBAC_MANIFEST[("flink-rbac.yaml\nRole + RoleBinding")]
    end

    %% ── make cp-up dependency chain
──────────────────────────────────────
    CP_UP --> CHECK_PRE
    CP_UP --> MK_START
    CP_UP --> CP_CORE_UP
    CP_UP --> KUI_INSTALL

    CP_CORE_UP["cp-core-up"] --> OP_INSTALL
    CP_CORE_UP --> CP_DEPLOY
    OP_INSTALL --> NS

    %% ── make flink-up dependency chain
──────────────────────────────────────
    FLINK_UP --> CERT
    FLINK_UP --> FL_OP
    FLINK_UP --> CMF_INSTALL
    FLINK_UP --> CMF_ENV
    FLINK_UP --> FL_DEPLOY
    FL_OP --> NS
```

```
    FL_DEPLOY --> FL_RBAC
    FL_DEPLOY --> MANIFEST
    FL_RBAC --> RBAC_MANIFEST

    %% ── make deploy-cp-ptf-udf dependency chain
────────────────────────
    DEPLOY_CP_PTF --> BUILD_PTF

    %% ── make deploy-cc-ptf-udf dependency chain
────────────────────────
    DEPLOY_CC_PTF --> BUILD_PTF

    %% ── make cp-down dependency chain
──────────────────────────────────
    CP_DOWN --> KUI_UN
    CP_DOWN --> CP_DELETE
    CP_DOWN --> OP_UNINSTALL

    %% ── make flink-down dependency chain
────────────────────────────────
    FLINK_DOWN --> FL_DELETE
    FLINK_DOWN --> CMF_UN
    FLINK_DOWN --> FL_OP_UN
    FLINK_DOWN --> CERT_UN

    %% ── make confluent-teardown
────────────────────────────────────────
    TEARDOWN -->|"1 - check minikube running"| FLINK_DOWN
    TEARDOWN -->|"2"| CP_DOWN
    TEARDOWN -->|"3 - delete namespace"| NS
    TEARDOWN -->|"4"| MK_STOP

    %% ── UI access
──────────────────────────────────────────────────────
    CP_DEPLOY -.->|"once Running"| C3
    FL_DEPLOY -.->|"once Running"| FL_UI
    CMF_INSTALL -.->|"once Running"| CMF_OPEN
    CMF_INSTALL -.->|"C3 Flink tab"| CMF_PROXY
    KUI_INSTALL -.->|"once Running"| KUI_OPEN

    %% ── Styles
──────────────────────────────────────────────────────────
    classDef entry    fill:#1a1a2e,stroke:#e94560,color:#fff,font-
weight:bold
    classDef install  fill:#16213e,stroke:#0f3460,color:#a8dadc
    classDef remove   fill:#2d1b1b,stroke:#8b0000,color:#ffb3b3
    classDef ui       fill:#1b2d1b,stroke:#2d6a2d,color:#b3ffb3
    classDef file     fill:#2d2b1b,stroke:#8b7500,color:#ffe680
    classDef composite fill:#2a1a2e,stroke:#9b59b6,color:#dbb8ff

    class CP_UP,FLINK_UP,CP_DOWN,FLINK_DOWN,TEARDOWN entry
    class CP_CORE_UP,DEPLOY_CP_PTF,DEPLOY_CC_PTF composite
    class
CHECK_PRE,MK_START,NS,OP_INSTALL,CP_DEPLOY,CERT,FL_OP,FL_RBAC,FL_DEPLOY,CM
```

```
F_INSTALL,CMF_ENV,CMF_PROXY,KUI_INSTALL,BUILD_PTF install
    class
MK_STOP,OP_UNINSTALL,CP_DELETE,FL_DELETE,FL_OP_UN,CERT_UN,CMF_UN,KUI_UN,TE
ARDOWN_CP_PTF,TEARDOWN_CC_PTF remove
    class C3,FL_UI,CMF_OPEN,KUI_OPEN ui
    class MANIFEST,RBAC_MANIFEST file
```

1.1.4 Using the `Makefile` Targets Quickstart

1.1.4.1 Full stack (CP + Kafka UI)

```
make cp-up
```

This runs: `check-prereqs` → `minikube-start` → `namespace` → `operator-install` → `cp-deploy` → `kafka-ui-install`.

Once pods are up, open Control Center:

```
make c3-open        # http://localhost:9021
```

1.1.4.2 Add Apache Flink + CMF (run separately after `make cp-up`)

```
make flink-up
```

This runs: `flink-cert-manager` → `flink-operator-install` → `cmf-install` → `cmf-env-create` → `flink-deploy`. `flink-up` is self-contained and can also be run standalone on a fresh cluster.

Once the Flink JobManager pod is running:

```
make flink-ui       # http://localhost:8081 (background — returns prompt)
make flink-ui-stop  # stop the background port-forward
make cmf-open       # http://localhost:8080/cmf/api/v1/environments
```

To expose the Flink tab inside Control Center, inject the CMF proxy sidecar:

```
make cmf-proxy-inject
```

1.1.5 `Makefile` Composite Workflow Target Reference

Target	What it does
`make cp-up`	Full stack: Minikube + CP + Kafka UI
`make flink-up`	cert-manager + Confluent Flink Operator + CMF + Flink cluster
`make cp-down`	Remove CP, Kafka UI, and Operator (Minikube keeps running)
`make flink-down`	Remove Flink cluster, CMF, Operator, and cert-manager
`make confluent-teardown`	Full teardown: everything + stop Minikube

1.1.6 `Makefile` Individual Target Reference

▶ Phase 1 — Prerequisites

Target	Description
`install-prereqs`	Install Docker Desktop, kubectl, Minikube, Helm, envsubst, and Gradle via Homebrew
`check-prereqs`	Verify all required tools are available

▶ Phase 2 — Minikube

Target	Description
`minikube-start`	Start Minikube with configured resources
`minikube-status`	Show Minikube and node status
`minikube-stop`	Stop the Minikube cluster
`minikube-delete`	Permanently delete the Minikube cluster

▶ Phase 3 — Confluent Operator

Target	Description
`namespace`	Create the `confluent` namespace and set it as default context
`operator-install`	Add Confluent Helm repo and install CFK Operator
`operator-status`	Show CFK Operator pod status
`operator-uninstall`	Remove the CFK Operator Helm release

▶ Phase 4 — Confluent Platform

Target	Description
`cp-deploy`	Deploy Kafka (KRaft), Schema Registry, Connect, ksqlDB, REST Proxy, Control Center
`cp-watch`	Watch pod startup live (Ctrl+C to exit)
`cp-status`	Show current pod status
`cp-delete`	Remove all CP components and leftover PVCs

▶ Phase 5 — Control Center

Target	Description

Target	Description
`c3-open`	Port-forward Control Center in the background and open `http://localhost:9021` (`make c3-stop` to kill)
`c3-stop`	Stop the background Control Center port-forward

▶ Phase 6 — Apache Flink

Target	Description
`flink-cert-manager`	Install cert-manager (Confluent Flink Operator dependency)
`flink-operator-install`	Install the Confluent Flink Kubernetes Operator (`confluentinc/flink-kubernetes-operator`)
`flink-operator-status`	Show Flink Operator pod status
`flink-operator-uninstall`	Remove the Confluent Flink Operator Helm release
`flink-rbac`	Apply supplemental RBAC so the `flink` SA can read services (needed for job submission)
`flink-deploy`	Deploy the Flink session cluster using `FLINK_MANIFEST` (runs `flink-rbac` first)
`flink-status`	Show Flink pods and FlinkDeployment CRs
`flink-ui`	Port-forward Flink UI in the background and open `http://localhost:8081` (`make flink-ui-stop` to kill)
`flink-ui-stop`	Stop the background Flink UI port-forward
`flink-delete`	Delete the Flink session cluster
`cert-manager-uninstall`	Remove cert-manager

▶ Phase 7 — Confluent Manager for Apache Flink (CMF)

Target	Description
`cmf-install`	Install CMF via Helm (`confluent-manager-for-apache-flink`) and wait for pod readiness
`cmf-env-create`	Create a Flink environment (`CMF_ENV_NAME`) in CMF pointing to the `confluent` namespace
`cmf-status`	Show CMF pod status and list registered Flink environments
`cmf-open`	Port-forward CMF REST API and open `http://localhost:8080/cmf/api/v1/environments`
`cmf-uninstall`	Uninstall CMF (safe to run even if not installed)

Target	Description
`cmf-proxy-inject`	Patch the C3 StatefulSet with a `socat` sidecar to expose the Flink tab in Control Center
`cmf-proxy-remove`	Remove the CMF proxy sidecar and resume CFK reconciliation
`cmf-proxy-logs`	Stream logs from the `cmf-proxy` sidecar in the C3 pod

▶ Phase 8 — Kafka UI (Provectus)

Target	Description
`kafka-ui-install`	Install Kafka UI connected to the local CP cluster (Kafka + Schema Registry + Connect)
`kafka-ui-status`	Show Kafka UI pod status
`kafka-ui-open`	Port-forward Kafka UI and open `http://localhost:8080`
`kafka-ui-uninstall`	Remove Kafka UI

▶ Phase 9 — Build & Deploy Flink JARs

Target	Description
`build-ptf-udf`	Build the `ptf_udf` fat JAR (requires Gradle)
`deploy-cp-ptf-udf`	Build UDF JAR, copy to Flink pods, and submit SQL via CMF
`teardown-cp-ptf-udf`	Tear down the SQL-based ptf_udf deployment via CMF

1.1.7.0 `Makefile` Target Configuration Reference

All variables are overridable at the command line.

▶ Defaults

Variable	Default	Description
`NAMESPACE`	`confluent`	Kubernetes namespace
`CONFLUENT_MANIFEST`	`k8s/base/confluent-platform-c3++.yaml`	Path to Confluent Platform manifest
`MINIKUBE_CPUS`	`6`	vCPUs allocated to Minikube
`MINIKUBE_MEM`	`20480`	Memory in MB
`MINIKUBE_DISK`	`50g`	Disk size
`FLINK_OPERATOR_VER`	`1.130.0`	Confluent Flink Kubernetes Operator version

Variable	Default	Description
`FLINK_IMAGE`	`confluentinc/cp-flink:2.1.1-cp1-java21-arm64`	Flink container image
`FLINK_VERSION`	`v2_1`	Flink API version string for the FlinkDeployment CR
`FLINK_CLUSTER_NAME`	`flink-basic`	Name of the FlinkDeployment resource
`FLINK_MANIFEST`	`k8s/base/flink-basic-deployment.yaml`	Path to FlinkDeployment template
`FLINK_RBAC_MANIFEST`	`k8s/base/flink-rbac.yaml`	Path to supplemental RBAC manifest for the `flink` ServiceAccount
`CERT_MANAGER_VER`	`v1.18.2`	cert-manager version
`CMF_VER`	`2.1.0`	Confluent Manager for Apache Flink version
`CMF_PORT`	`8080`	CMF REST API local port
`CMF_ENV_NAME`	`dev-local`	Flink environment name registered in CMF
`C3_PORT`	`9021`	Control Center local port
`FLINK_UI_PORT`	`8081`	Flink UI local port
`KAFKA_UI_PORT`	`8080`	Kafka UI local port
`PTF_UDF_TOPICS`	`user_events enriched_events`	Kafka topics for the ptf_udf Flink job

> **Note:** CMF uses the Confluent-packaged Flink operator (`confluentinc/flink-kubernetes-operator`) and `confluentinc/cp-flink` images — not the Apache OSS Flink operator or `flink` Docker Hub image.

Example — deploy a specific Flink image:

```
make flink-deploy FLINK_IMAGE=confluentinc/cp-flink:2.1.1-cp1-java21-arm64
FLINK_VERSION=v2_1
```

1.1.8 `Makefile` Target Teardown

Remove everything and stop Minikube:

```
make confluent-teardown
```

To keep Minikube running but remove all deployed components:

```
make flink-down   # Flink cluster + CMF + operator + cert-manager
make cp-down      # CP + Kafka UI + CFK Operator
```

1.2 Confluent Cloud Setup (early access example)

1.2.1 Required Tools

Before you begin, ensure you have access to the following cloud accounts:

- **Confluent Cloud Account** — for Kafka and Schema Registry resources
- **Terraform Cloud Account** — for automated infrastructure provisioning

Make sure the following tools are installed on your local machine:

- **Confluent CLI version 4 or higher**
- **Java JDK 17**
- **Terraform CLI version 1.13.0 or higher**

2.0 The Examples

Once the platform is up, head to the examples:

Example Type	Example Description	Confluent Platform + Minikube	Confluent Cloud
PTF UDF-type (state-driven)	Walks through building, deploying, and testing a stateful **ProcessTableFunction** that enriches Kafka user events with per-user session tracking.	`CP Deploy`	`CC Deploy`

3.0 Debugging a Flink UDF

You can attach your IDE's debugger (VS Code or IntelliJ IDEA) to a running Flink TaskManager and *hit breakpoints inside your UDF code* — even though it's executing on a remote Java Virtual Machine (JVM) inside Kubernetes. The `FlinkDeployment` Custom Resource (CR) already has **Java Debug Wire Protocol (JDWP)** enabled, and debug configurations are pre-wired for both VS Code and IntelliJ IDEA.

Prerequisites: The Confluent Platform and Flink stack must be running (`make cp-up && make flink-up`), and your UDF must be deployed (`make deploy-cp-ptf-udf`).

1. **Set a breakpoint** — open `UserEventEnricher.java` and click in the gutter at the first line of the `eval()` method (line 92):

```
String eventType = input.getFieldAs("event_type");
```

2. **Attach the debugger** — select the **"Attach to Flink TaskManager"** configuration and start debugging. The IDE will automatically port-forward to the TaskManager pod and attach to the JDWP agent on port `5005`.
   - **VS Code:** Open the **Run and Debug** panel (⇧⌘D), select the configuration from the dropdown, and press **F5**
   - **IntelliJ IDEA:** Open the **Run/Debug Configurations** dropdown (top-right toolbar), select the configuration, and click **Debug** (^D / Shift+F9)
3. **Send a test event** — produce a single JSON message to the `user_events` topic to trigger the breakpoint:

   ```
   make produce-user-events-record
   ```

4. **Debug** — your IDE will pause at your breakpoint. You can inspect `input`, `state`, and local variables, step through the session logic, and watch `state.sessionId` and `state.eventCount` update as you step over lines.

> For the full deep-dive (how JDWP is configured, how port-forwarding works, important caveats), see Remote Debugging a Flink PTF UDF.

4.0 Resources

4.1 Confluent for Kubernetes (CfK)

- Manage Confluent Platform with Confluent for Kubernetes
- Minikube

4.2 Confluent Platform for Apache Flink

- Stream Processing with Confluent Platform for Apache Flink
- Architecture and Features of Confluent Platform for Apache Flink
- Get Started with Confluent Platform for Apache Flink

4.3 Confluent Cloud for Apache Flink (CCAF)

- Stream Processing with Confluent Cloud for Apache Flink
- Get Started with Confluent Cloud for Apache Flink